Republic of the Philippines
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LAB REPORT

(LINEAR SEARCH ALGORITHM)

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MACHINE PROBLEM DESCRIPTION:

This program implements and analyzes two essential searching algorithms:

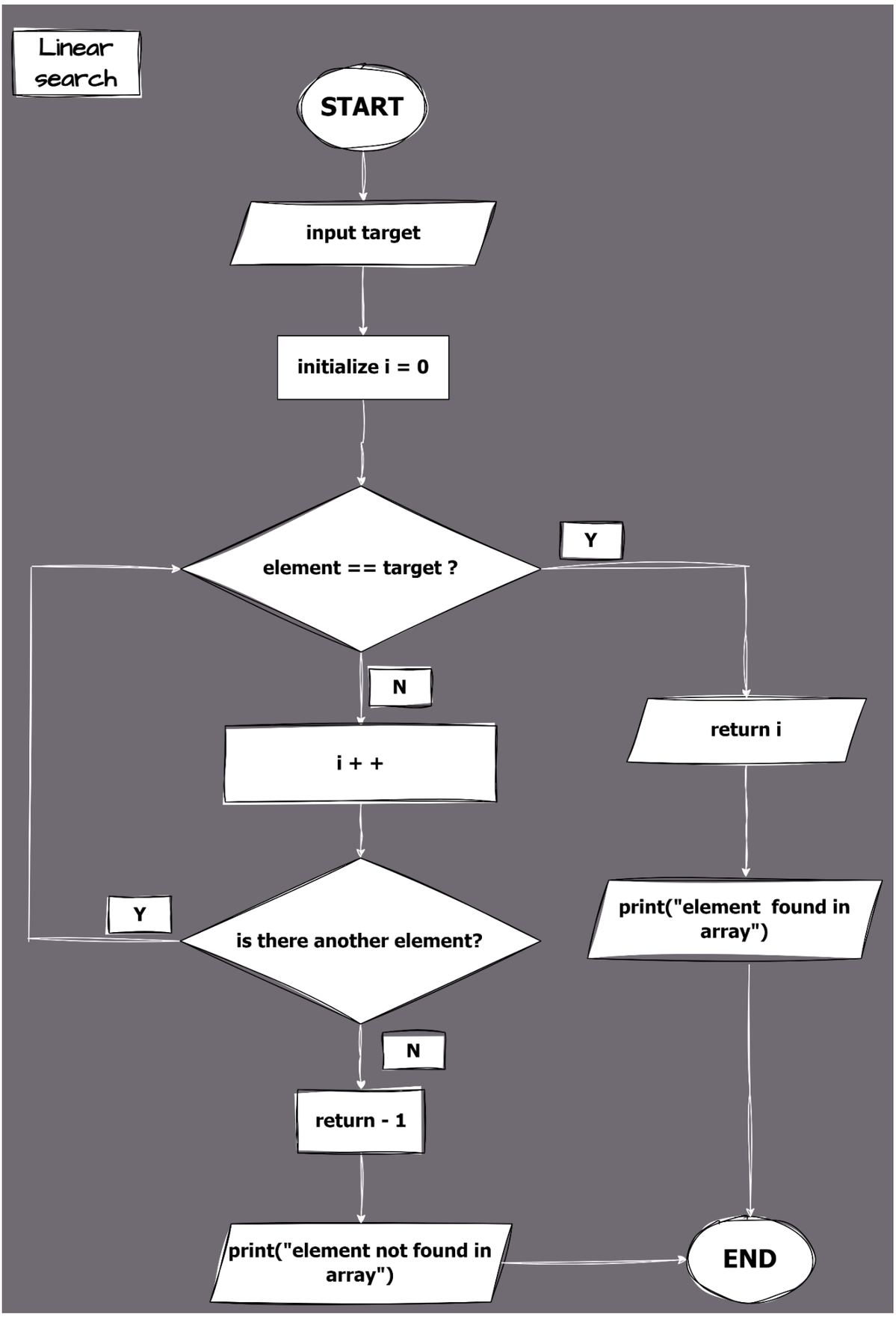
LINEAR SEARCH:

This algorithm employs a sequential approach, systematically examining each element of a list to identify a target value. It ensures thoroughness and simplicity, making it suitable for small to medium-sized datasets.

Through this program, we aim to explore the characteristics, advantages, and applications of each algorithm, gaining a deeper understanding of their roles in solving real-world search problems.

FLOWCHART (LINEAR SEARCH):

This flowchart represents a scenario where the program processes an array using the Linear Search algorithm. The flowchart details the step-by-step process:



SOURCE CODE:

The code below demonstrates the application of the Linear Search algorithm on an array of integers.

LinearSearch.py

```python
def linearsearch(arr, target):
    for i, element in enumerate(arr):
        if element == target:
            return i
    return -1

if __name__ == "__main__":
    arr = [65, 49, 10, 2, 43, 83, 89, 79, 9, 24, 12, 80, 81, 53, 68, 4]
    target = int(input("Enter the key to search: "))
    result = linearsearch(arr, target)

    if result != -1:
        print(f"Element found at index {result}")
    else:
        print("Element not found in the array")
```

OUTPUT:

The output displays whether the target value was found in the array and at which index. If the target value is not present, the output will state, "Element not found in the array."

Element Found:

This output shows the case where the target element is present in the array.

```
Run    LinearSearch
"C:\Users\Gian Carlo Trilles\PycharmProjects\pythonProject\.venv\Scripts\python.exe"
Enter the key to search: 68
Element found at index 14

Process finished with exit code 0
```

Element Not Found:

This output shows the case where the target element is not present in the array.

```
Run    LinearSearch
"C:\Users\Gian Carlo Trilles\PycharmProjects\pythonProject\.venv\Scripts\python.exe"
Enter the key to search: 5
Element not found in the array

Process finished with exit code 0
```